

Cognitive Ability Classification using On-body Sensors

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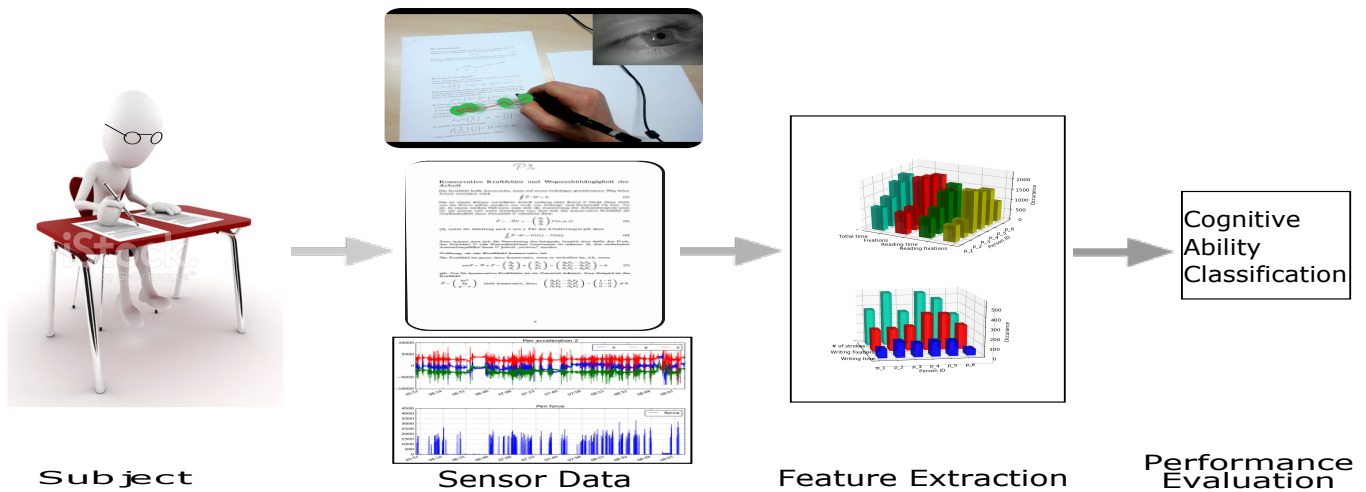


Figure 1: Cognitive ability classification process

ABSTRACT

This work presents an on-body sensor setup to keep track of the learning progress while performing cognitive activities in formal education. The proposed approach uses the sensor collected data for cognitive ability classification, i.e., confidence score, cognitive load, and expertise, while attempting to solve physics-related problems. Data is collected using a combination of instruction material, an eye-tracker, and a sensor pen to capture the cognition progress. The collected data is processed and transformed to feature representation using a novel feature-set presented in this paper for learning analytics. The used features help to highlight the difference in learner's cognitive abilities and enable teachers to cater to individual needs and requirements, an essential aspect of need-based learning.

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CCS CONCEPTS

• **Human-centered computing** → **Visualization design and evaluation methods**; • **Computer systems organization** → **Embedded systems**; • **Applied computing** → **Interactive learning environments**;

KEYWORDS

on-body sensors, cognitive ability classification, gaze detection, learning analytics, formal education, need-based learning

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1 INTRODUCTION

One of the main objectives of formal education is to instigate the curiosity to learn new things and stimulate the cognitive abilities of the students to develop problem-solving skills. *Cognitive abilities* are defined as the capacity of an individual: "to learn, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience"[16]. The learning habits of every individual vary based on their cognitive abilities, interests, and preferences. On the other side of the learning process, instructors

play a vital role in steering the cognitive potential in the learning contexts by deciding what to teach and how to teach it. It's very hard for a teacher to track and monitor the progress of every student individually and address them using traditional teaching approaches. On-body sensor setup for learning analytics can act as a bridge in the student-teacher relationship by assisting in performing and tracking the activities, evaluating behaviours, and performance effectively.

In the recent past, significant developments have been made to integrate technological developments and smart gadgets in formal education to strengthen student-teacher interaction bonding [8, 12]. Using Sensors for effective interpretation of cognitive activities is a common practice nowadays [2, 3, 15]. The concept of intelligent and interactive documents has been presented recently, where dynamic handling of documents results in better engagement and focus during the reading activity [10]. Gaze-tracking methods are used to understand visual attention for STEM education [12], attention analysis using different modes of instruction [18], and visual attention while performing Physics experimentation [13]. Another study [7] uses video analysis to evaluate the cognitive load for multiple representation learning. Using implicit sensor information for cognitive ability classification to foster representational competence for teachers and learners is still an open research area to the best of the author's knowledge.

The presented work focuses on cognitive ability classification using a sensor-based learning analytic platform. The primary motivation behind our work is using a combination of on-body sensors to analyze the cognitive activities in the classroom, such as reading, writing, and problem-solving. Initial data exploration results regarding confidence score, cognitive load, and expertise are discussed to look into the behaviours of experts and novices. The insights into the behaviours signify the differences in cognitive abilities of both groups using the proposed feature set. Teachers can use the derived insights to indicate the weaknesses and strengths of an individual student to help them give feedback based on their cognitive and effective requirements.

2 SYSTEM OVERVIEW

Research studies have established that aiding human activities with technological interventions positively impacts performance and outcomes [6]. Our presented system is centred around the students aided with on-body sensors, i.e., eye tracker, sensor pen, and meta information, to track performance while performing cognitive activities. The complete system with its components is shown in figure Figure 1.

2.1 Meta information

Multiple representations help people in effective learning of complex ideas [1]. Meta information defines the structure of documents and includes instructional material and a set of exercises to be worked out by participants. Instruction material also includes the reading text to familiarise the participants with the relevant information and concept of gravitational field theory. Exercises involve the complex representation learning to analyze the progress of students while learning physical content.

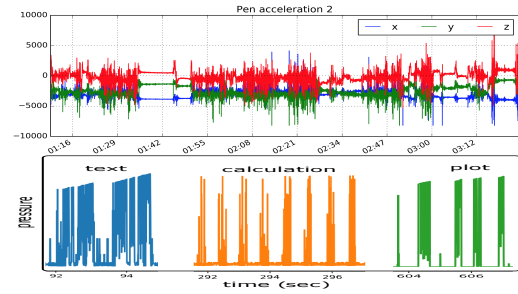


Figure 2: Pen sensors data showing raw information signal from IMU and pen pressure signal while writing



Figure 3: Eye-tracker in application for gaze-tracking along with eye view

2.2 Digital pen

Sensor pens and graphics tablets are now standard means to capture the handwriting progress for digitization. Handwriting features play a vital role in the evaluation of cognitive load [22, 23]. We used a Stabilo digi-pen¹, a sensor pen with an IMU sensor and a pressure sensor. IMU capture data tracks the pen movement, a vital step in reconstructing the writing progress. Pressure signal records the pen-up and pen-down activity while writing. Figure 2 shows the raw IMU & pressure sensor data while producing complex representation, i.e., normal text, equations or drawing figures.

2.3 Eye-tracker

Reading behaviour is a very important aspect of cognitive ability classification [17]. Capturing eye movements to understand behaviour while performing cognitive activities is promising, as the eyes play a vital role during these activities [5, 9]. In this study, Pupil monocular, wearable mobile eye-tracker is used [11]. Pupil eye-tracker captures both eye movements and real-world camera view. It can also detect pupil, track pupil movement, maps gaze data in real-time, and many other functions. Gaze mappings and pupil eye-tracker are shown in Figure 3.

3 EXPERIMENTATION

3.1 Data Collection

Six participants (4 males, 2 females) volunteered for the study. Out of six participants, 2 were students, and 4 were researchers from the

¹<https://stabilodigital.com/digipen/>

Physics department. Before attempting solutions, three participants were categorized as novices and the rest as experts based on their understanding and self-assessment of the topic.

Participants were presented with instructional material containing reading material and a set of tasks for problem-solving related to gravitational field theory. Exercises were designed to test factual knowledge and knowledge transfer skills. Solutions to the exercises required hand-written descriptive answers, mathematical calculations, and complex representations such as drawing figures and plots.

3.2 Feature Extraction

We present a novel set of 12 features to analyse the behaviours for cognitive ability classification. Pen sensor data are used to extract the features related to writing activity. Eye-trackers data are used to extract the eye-gaze information while performing the reading and writing activity or both.

- **Total fixations**, total number of fixations occurred for a participant to complete the task.
- **Reading fixations**, fixations occurred while reading or thinking process during the whole activity.
- **Writing fixations**, fixations happened only during the writing activity.
- **Average consecutive read fixations**, average fixations count occurred for a single reading or cognition before switching to writing.
- **Average consecutive write fixations**, average number of consecutive fixations for a person during the writing activity.
- **Total time**, time taken in seconds by a participant to complete the whole activity including reading, writing, and cognition.
- **Reading time**, time in seconds consumed during reading activity, cognition, and understanding the problem.
- **Writing time**, time in seconds consumed by a participant for writing activity.
- **Sketched segments**, count of written segments produced as text, mathematical calculation, plots and diagrams during the activity period.
- **Shift between read & write**, number of times a participant was engaged in cognitive process or consulting instruction material while producing solutions.
- **Average time difference between strokes**, average time taken between two consecutive pen usages.
- **Writing pressure**, one of the key factor to estimate the stress while performing writing activity.

4 INITIAL DATA EXPLORATION

Tchalenko et al. [19] present a study on an "eye-hand" strategy to copy and produce drawings for experts and novices. Alamargot et al. [2] presented an "Eye and Pen" method for looking into the dynamics of the writing process based on visual attention. We employ the proposed feature set to evaluate its effectiveness and relevance in cognitive ability classification. Figure 4 highlights observable differences in behaviours of experts and novices for cognitive activities.

Confidence score. Self-confidence is an essential factor in learning. Feedback based on self-confidence during cognitive activities helps the learners improve performance [14]. Figure 4c shows that experts need lesser time to think while attempting solutions. They also provide precise and compact solutions to the problems, indicating higher self-confidence. Similarly, Figure 4c provides valuable insights about initial cognition, as novices require not only more time to complete the tasks but also to understand the task, as compared to experts.

Cognitive load. Hochberg et al. [7] conducted a study to demonstrate that lower cognitive load results in higher learning achievements. Figure 4a shows that the novices need more time between pen usage. It means they experienced a higher cognitive load for attempting solutions to exercises. It is also observed that the novices consulted more often to information material while attempting solutions to exercises, as shown in Figure 4b. It also indicates that they had more difficulty in recalling the necessary information. Figure 4c depicts that the novices required significantly higher time for initial reading than experts, indicating a higher cognitive load while understanding the concepts and tasks.

Expertise. is defined as one's understanding and knowledge in a particular field. Brueckner et al. [4] explored the influence of expertise on representation learning across multiple domains in a recent study. In this work, we also explore the difference in behaviours of novices and experts based on their expertise. Figure 4 shows different metrics with notable differences in the behaviours of experts and novices in attempting solutions, recalling concepts, required lesser time to understand and produce solutions. Figure 4d shows the visible difference in results of experts and novices, even when the normalised sum of all features in the feature set is used.

In this study, we present a feature set to explore the difference between the behaviour of experts and novices when they are exposed to factual and knowledge transfer-based exercises. The proposed features give teachers meaningful insights about differences in behaviours for analysis, understanding, and approach to attempting tasks for experts and novices. Experts exhibit different behaviour from the beginning, whether analysing the problem, understanding the problem and formulating the solutions, taking less time, producing quick answers, skipping intermediate steps and producing abstract solutions. It is also observed that novices require more effort, time, and higher cognitive load from analyzing the task to translating their understanding on paper. Novices go through details and process every intermediate step on paper. These insights are helpful for both learners and teachers to analyse the performance based on an individual's strengths and weaknesses and address them accordingly, which helps improve the overall learning process.

5 CONCLUSION AND FUTURE WORK

On-body sensors enable the teachers to look deeper into the behaviours of the students, assist them in interacting and addressing the individual's requirements, and foster the concept of need-based learning. Initial data exploration reveals that implicit sensor information can be used as an aid for the teachers to provide needs-based individual feedback. The encouraging results and findings

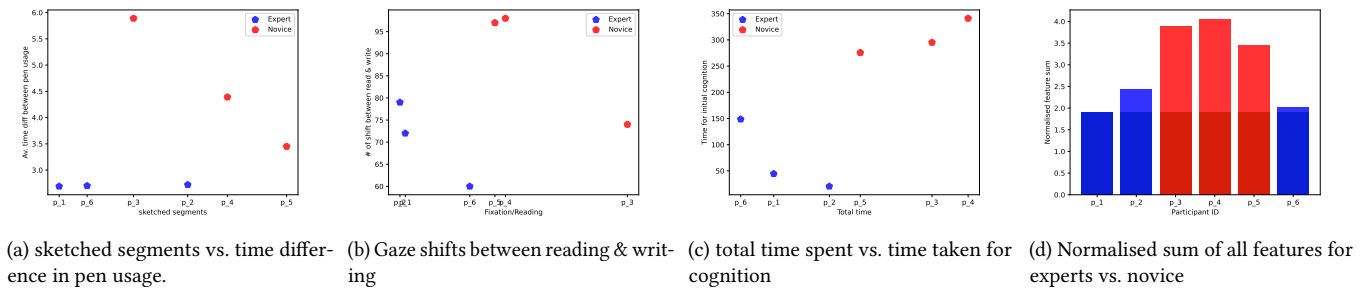


Figure 4: Cognitive ability comparison for experts and novices.

demand further research to develop mental models using on-body sensors for adaptive teaching and learning analytic systems. For future work, we recommend using artificial intelligence methods to process the sensor data for cognitive ability classification. We recommend incorporating the handwriting classifiers [20] and content classifier [21] for deeper insights into respective cognitive activities. These enhancements will help achieve the goal of adaptive and personalised need-based learning.

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